

EDUCATION

The Chinese University of Hong Kong

Ph.D. in Electronic Engineering Supervisor: Ren Hongliang

09 2023 – now

HK, ShaTin

Hong Kong University of Science and Technology

Beng in Computer Science and Engineering

09 2019 – 06 2023

HK, Clear Water Bay

PROJECTS

3D Perception for Surgical Robotics

Supervisor: Ren Hongliang (CUHK)

06 2023 - Present

- Dynamic Endoscopic Robotics 3D Perception.
- 3DGS-based VLM & MLLM Surgical Scene Understanding.
- Dynamic Surgical Scene Simulation

Driving World Action Model & LiDAR-assisted 3D Reconstruction

Supervisor: Weihua Zhang, Xin Kang (NVIDIA)

10 2025 - Present

- Cosmos-based World Model for Autonomous Driving VLA.
- Extrapolated View LiDAR Rendering Enhancement with 3DGS.

Multi-Modal Sensor-Agnostic Depth Enhancement

Supervisor: Chen Qifeng, Lei Chenyang (HKUST)

03 2023 - 12 2023

- Refine 3D for multiple sensors (Stereo, LiDAR, iToF, dToF) with RGB, Polarization, Sparse Depth.

SELECTED PUBLICATIONS

1. **Huang Y.**, et al. LiDAR-EVS: Enhance Extrapolated View Synthesis for 3D Gaussian Splatting with Pseudo-LiDAR Supervision. (ECCV 2026).
2. **Huang Y.**, et al. SurgTPGS: Semantic 3D Surgical Scene Understanding with Open-Vocabulary Gaussian Splatting. (MICCAI, 2025, **Young Scientist Award** (5/3677)).
3. **Huang Y.**, et al. Endo4DGX: Robust Endoscopic Scene Reconstruction and Illumination Correction with Gaussian Splatting. (MICCAI, 2025).
4. **Huang Y.**, et al. Advancing Dense Endoscopic Reconstruction with Gaussian Splatting-driven Surface Normal-aware Tracking and Mapping. (ICRA, 2025).
5. **Huang, Y.**, et al. Endo-4DGS: Endoscopic monocular scene reconstruction with 4D Gaussian Splatting. (MICCAI 2024).
6. Liu C.*, **Huang, Y.***, et al. EndoGSim: Physics-Aware 4D Dynamic Endoscopic Scene Simulations via MLLM-Guided Gaussian Splatting. (MICCAI 2026)
7. Cui B.*, **Huang, Y.***, et al. TR2M: Transferring Monocular Relative Depth to Metric Depth with Language Descriptions and Dual-Level Scale-Oriented Contrast. (CVPR 2026)
8. Lin Y.*, **Huang, Y.***, et al. EndoDDC: Learning Sparse to Dense Reconstruction for Endoscopic Robotic Navigation via Diffusion Depth Completion. (ICRA 2026)
9. Ikemura, K.*, **Huang, Y.***, et al. Robust Depth Enhancement via Polarization Prompt Fusion Tuning. (CVPR 2024).

Experience

NVIDIA HK. AI Research Intern

LiDAR, 3DGS, Feed-Forward Reconstruction, World Model

10 2025 – now

SenseTime HK. Computer Vision Intern

Low Level Vision, ISP, Object Detection

01 2022 – 08 2022

Enterprize (HKUST Robomaster) Infantry Robot Devel. Lead

Robot Design, Solidworks, Robot Manufacture

09 2020 – 08 2022